

Type I and II errors

Exercise 1. A city has introduced traffic calming measures, and if they don't work it will consider widening a busy intersection in town. Last year, the city reported 17,000 cars passed through the intersection per day. The city conducted a survey for 49 days this year and found an average of 16,000 cars passed through the intersection, with a sample standard deviation of 5000. Therefore, the population average for last year was 17,000 cars per day and the average for this year based on a sample of 49 days is 16,000 cars per day.

- Specify the null and alternative hypotheses to determine whether the intersection has seen a decrease in traffic this year.
- Calculate the value of the test statistic and approximate the p-value.
- The city is going to widen the intersection if it believes traffic has increased. At the 5% significance level, can you conclude that the intersection has seen an increase in traffic? Should the city widen the intersection?

Exercise 2. A portfolio manager claims that the mean annual return on one of the mutual funds he manages exceeds 8%. To substantiate his claim, he states that over the past 10 years, the mean annual return for the mutual fund has been 9.5% with a sample standard deviation of 1.5%. Assume annual returns are normally distributed.

- Specify the competing hypotheses to test the portfolio manager's claim.
- Calculate the value of the test statistic.
- At the 5% significance level, use the critical value approach to state the decision rule.
- Is the portfolio manager's claim substantiated by the data? Explain.

Exercise 3. From 2017/18 exam:

An economist is interested in the effects of exchange rate regimes on economic performance. She gathers the observations in the table below.

	Sample mean		Sample standard deviation		Sample size
Countries with:	inflation rate (%)	output growth (%)	inflation rate	output growth	
fixed exchange rates	8.4	3.1	2.01	0.46	532
floating exchange rates	15.2	2.9	2.27	0.35	219

Test the hypothesis that the mean inflation rate of countries with floating exchange rates is 10%.

- What is the appropriate null and alternative hypothesis?
- Calculate the test statistic. At a 5% significance level what do you conclude?
- What is the critical value at 5% significance level?
- What is your conclusion?

Exercise 4. A researcher plans to conduct a test of hypotheses at the 1% significance level. She designs her study to have a power of 0.90 at a particular alternative value of the parameter of interest.

- What is the probability that the researcher will commit a Type I error?

A) 0.01
B) 0.10
C) 0.90
D) 0.99

- What is the probability that the researcher will commit a Type II error for the particular alternative value of the parameter at which she computed the power?

A) 0.01
B) 0.10
C) 0.90
D) 0.99

Exercise 5. The nicotine content in cigarettes of a certain brand is Normally distributed with standard deviation $\sigma = 0.1$ milligrams. The brand advertises that the mean nicotine content of their cigarettes is $\mu = 1.5$, but you are suspicious and plan to investigate the advertised claim by testing the hypotheses $H_0 : \mu = 1.5$ versus $H_a : \mu > 1.5$ at the 5% significance level. You will do so by measuring the nicotine content of 15 randomly selected cigarettes of this brand and computing the mean nicotine content $\bar{x}$ of your measurements.

- What is the smallest value of $\bar{x}$ for which you will reject H_0 ?

A) 1.458
B) 1.505
C) 1.542
D) 1.551

b) If the mean nicotine content of the cigarettes is, in fact, $\mu = 1.6$, what is the power of the test?

- A) 0.7995
- B) 0.95
- C) 0.9711
- D) 0.9877

Exercise 6. The time needed for college students to complete a certain paper-and-pencil maze follows a Normal distribution with a mean of 30 seconds and a standard deviation of 3 seconds. You wish to see if the mean time μ is changed by vigorous exercise, so you have a group of nine college students exercise vigorously for 30 minutes and then complete the maze. Assume that σ remains unchanged at 3 seconds. You decide to test the hypotheses $H_0 : \mu = 30$ versus $H_a : \mu \neq 30$ at the 1% significance level.

a) What is the range of $\bar{x}$ -values for which you will *not* reject H_0 ?

- A) (27.424, 32.576)
- B) (27.674, 32.326)
- C) (29.141, 30.859)
- D) (29.225, 30.775)

b) What is the power of your test at $\mu = 28$ seconds?

- A) 0.8925
- B) 0.8752
- C) 0.3722
- D) 0.2823

Exercise 7. The attention span of little kids (ages 3–5) is claimed to be Normally distributed with a mean of 15 minutes and a standard deviation of 4 minutes. A test is to be performed to decide if the average attention span of these kids is really this short, or if it is longer. You decide to test the hypotheses $H_0 : \mu = 15$ versus $H_a : \mu > 15$ at the 5% significance level. A sample of 10 children will watch a

TV show they have never seen before, and the time until they walk away from the show will be recorded.

a) Fill in the blank. At a significance level of 5%, the decision rule would be to reject the null hypothesis if the observed sample mean is greater than _____ minutes.

- A) 15.66
- B) 17.08
- C) 17.48
- D) 19

b) If, in fact, the true mean attention span of these kids is 18 minutes, what is the probability of a Type II error?

- A) 0.0107
- B) 0.2335
- C) 0.3405
- D) 0.7665

Exercise 8. Errors: all questions use the following information:

Consider a problem with the following decision rule: Reject H_0 if $\frac{\bar{x}-4}{0.08 \cdot \sqrt{25}} > 4.026$. So the null is $\mu \leq 4.00$

1. Rewrite the decision rule explicitly in terms of $\bar{x}$ and compute the probability of a Type II error and the power of the test for $\mu \geq 4.06$
2. Rewrite the decision rule explicitly in terms of $\bar{x}$ and compute the probability of Type II error and the power of the test for $\mu \geq 4.04$
3. Rewrite the decision rule explicitly in terms of $\bar{x}$ and compute the probability of Type II error and the power of the test for $\mu \geq 4.01$
4. Rewrite the decision rule explicitly in terms of $\bar{x}$ and compute the probability of Type II error and the power of the test for $\mu \geq 4.001$